Transport Over Wireless

Content

Internet Protocol Suite

– Link Layer: Ethernet, PPP, ARP, MAC Addressing

– Network Layer: IP, ICMP, Routing

– Transport Layer: TCP, UDP, Port Numbers, Sockets

– Application Layer: FTP, Telnet & Rlogin, HTTP, RTP

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Internet Protocol Suite:

1. **What is the purpose of ARP?**

   ARP (Address Resolution Protocol) is used to map IP addresses to MAC addresses in a local network. Its purpose is to enable communication between devices within the same network by resolving IP addresses to corresponding MAC addresses, which are used for data transmission at the data link layer.

2. **What is the difference of ARP from DNS?**

   ARP resolves IP addresses to MAC addresses within a local network, facilitating communication between devices. DNS (Domain Name System) resolves domain names to IP addresses globally, enabling devices to locate and communicate with servers and services across the internet.

3. **How does ARP differ from DHCP?**

   ARP is used for mapping IP addresses to MAC addresses within a local network, whereas DHCP (Dynamic Host Configuration Protocol) is used for dynamically assigning IP addresses and other network configuration parameters to devices on a network

4. **What is the purpose of encapsulation and decapsulation?**

   Encapsulation involves adding protocol headers to data packets as they move down the network stack, providing necessary information for transmission and routing. Decapsulation is the process of removing these headers as packets move up the network stack at the destination

5. **Why do we need multiple headers? Why not a single one?**

   Multiple headers are needed to provide various layers of functionality and abstraction in network communication. Each layer of the network stack adds its header containing relevant information for its specific purpose, allowing for modular and flexible communication protocols.

6. **What does the MTU mean? What is the purpose of the path MTU?**

   MTU (Maximum Transmission Unit) refers to the maximum size of a data packet that can be transmitted over a network link. Path MTU refers to the minimum MTU value along the path from the source to the destination. Its purpose is to prevent packet fragmentation and optimize data transmission by determining the maximum packet size that can be transmitted without fragmentation along the entire path.

7. **What is the purpose of the different header fields in IP? What is the purpose of the checksum and the protocol field?**

   IP header fields serve various purposes in the IP protocol. The checksum field ensures data integrity during transmission, while the protocol field specifies the higher-layer protocol (such as TCP or UDP) used for packet delivery.

8. **What is the purpose of ports used by TCP and UDP? How does it differ from IP addresses?**

   Ports are used by TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) to differentiate between multiple communication endpoints on a single device. They enable multiplexing of network traffic and facilitate communication between applications running on different devices. Unlike IP addresses, which identify devices, ports identify specific services or processes running on those devices.

9. **How are errors handled with UDP? Why is UDP not considered reliable?**

   UDP (User Datagram Protocol) does not provide error detection or correction mechanisms like TCP. Errors in UDP are handled at the application layer, where applications may implement their own error detection and recovery mechanisms. UDP is considered unreliable because it does not guarantee delivery or order of packets, making it suitable for applications where real-time performance is prioritized over reliability

10. **Why does the TCP/IP model provide different transport layer protocols?**

    The TCP/IP model provides different transport layer protocols (such as TCP and UDP) to accommodate different communication requirements. TCP offers reliable, connection-oriented communication with features like error detection, flow control, and sequencing, making it suitable for applications that require guaranteed delivery and data integrity. UDP, on the other hand, offers lightweight, connectionless communication with minimal overhead, making it suitable for applications that prioritize low latency and real-time performance over reliability.

TCP (Transport Control Protocol)

1. **How does the MSS differ from the MTU?**

   MSS (Maximum Segment Size) refers to the largest amount of data that can be sent in a single TCP segment, excluding the TCP header. MTU (Maximum Transmission Unit) refers to the largest size of a data packet that can be transmitted over a network link, including the IP and TCP headers. The MSS is typically smaller than the MTU to account for the additional overhead of headers.

2. **What is the purpose of the MSS?**

   The purpose of MSS is to optimize data transmission by specifying the largest segment size that can be sent without fragmentation. It helps prevent packet fragmentation and ensures efficient use of network resources by adjusting the segment size to match the capabilities of the network path

3. **What is the purpose of the different flags?**

   Flags in TCP header, such as SYN, ACK, FIN, RST, and URG, serve various purposes in controlling and managing TCP connections. For example, SYN is used for connection establishment, ACK acknowledges received data, FIN indicates the end of data transmission, RST resets the connection, and URG signals urgent data.

4. **What is the purpose of the checksum field?**

   The checksum field in the TCP header is used to ensure the integrity of the TCP segment during transmission. It calculates a checksum value based on the segment's contents and verifies it at the receiving end to detect any errors or corruption that may have occurred during transmission.

5. **What is the purpose of the window size field?**

   The window size field in the TCP header indicates the amount of data, in bytes, that a sender can transmit before requiring an acknowledgment from the receiver. It helps regulate the flow of data between sender and receiver, enabling efficient utilization of network resources and preventing congestion.

6. **What is the purpose of DUPACKs?**

   DUPACKs (Duplicate Acknowledgments) are used by TCP to detect packet loss and trigger fast retransmission. When a receiver receives out-of-order segments, it sends DUPACKs to acknowledge the highest sequentially received segment, indicating that the sender should retransmit the missing segment.

7. **How does a TCP sender learn about overload in the network? Which messages and mechanisms are used for this?**

   TCP sender learns about network overload through the observation of packet loss, increased round-trip times (RTTs), and the receipt of congestion signals such as Explicit Congestion Notification (ECN). These signals indicate network congestion, prompting the sender to adjust its transmission rate accordingly.

8. **How does the TCP congestion control algorithm in the sender react to overload problems? What are the options?**

   In response to network overload, TCP sender employs congestion control algorithms such as slow start, congestion avoidance, fast retransmit, and fast recovery. These algorithms dynamically adjust the sender's transmission rate, window size, and retransmission behavior to alleviate congestion and maintain network stability.

9. **Why does TCPs response to timeouts differ from the response for DUPACKs? What is the purpose of this?**

   TCP's response to timeouts and DUPACKs differs because they indicate different scenarios of packet loss. Timeouts occur when a sender does not receive an acknowledgment within a specified timeout period, suggesting severe network congestion or packet loss. DUPACKs, on the other hand, indicate minor packet loss or reordering in the network, triggering fast retransmission without waiting for a timeout.

10. **Can we find out the direction in which the congestion occurs?**

    TCP typically does not directly determine the direction of congestion. However, variations in round-trip times, packet loss, and congestion signals observed by the sender can provide indirect indications of congestion direction. Additionally, mechanisms like Explicit Congestion Notification (ECN) can help routers signal congestion, aiding in determining the direction of congestion.

TCP over Wireless

1. **What are the problems of the use of TCP over wireless links?**

   High Packet Loss: Wireless links are prone to packet loss due to factors like interference, fading, and signal attenuation, leading to degraded TCP performance.

   Variable and Unpredictable Latency: Wireless links often exhibit variable and unpredictable latency, which can affect TCP's congestion control mechanisms and lead to suboptimal performance.

   Limited Bandwidth: Wireless links typically have lower bandwidth compared to wired links, constraining TCP's throughput and efficiency

   Channel Errors and Corruption: Wireless links are susceptible to channel errors and corruption, which can result in retransmissions and impact TCP's reliability and performance.

2. **Identify possible effects of high fluctuations of the channel capacity on TCP?**

   Congestion: High fluctuations in channel capacity can lead to congestion and increased packet loss, triggering TCP's congestion control mechanisms and reducing throughput.

   Increased Retransmissions: Fluctuations in channel capacity may cause frequent changes in network conditions, leading to increased retransmissions and overhead for TCP.

   Unpredictable Performance: Variations in channel capacity can result in unpredictable performance for TCP, causing fluctuations in throughput, latency, and overall connection quality.

3. **What is the difference in the impact of temporarily poor radio channels for the case that LLC is empty or uses some ARQ over the wireless link?**

   Empty LLC: In the case of an empty LLC (Logical Link Control) layer without ARQ (Automatic Repeat reQuest), TCP relies solely on its built-in mechanisms such as retransmissions and congestion control to recover from packet loss. Temporarily poor radio channels can result in increased packet loss and retransmissions, leading to degraded TCP performance

   ARQ: When ARQ is used over the wireless link, it provides additional error detection and correction capabilities. Temporarily poor radio channels may still cause packet loss, but ARQ can help mitigate the impact by retransmitting lost packets, improving TCP's reliability and performance

4. **Can you think of mechanisms to alleviate the problem?**

   Forward Error Correction (FEC)
   Adaptive Retransmission Timeout (RTO)
   Selective ACKnowledgment (SACK)

Cross-Layer Optimization

5. **What is the advantage of splitting TCP connections in a wired and a wireless part, what's the disadvantage?**

   Advantage: Splitting TCP connections into wired and wireless parts allows for separate optimization and adaptation strategies tailored to the characteristics of each link. This can improve overall performance and reliability, especially in heterogeneous network environments

   Disadvantage: Splitting TCP connections introduces complexity and overhead in managing multiple connections, potentially leading to increased latency, resource consumption, and management overhead

6. **Are there less intrusive solutions to the TCP-over-wireless problem?**

   TCP Variants: Deploying TCP variants specifically designed for wireless networks, such as TCP Westwood or TCP Vegas, which incorporate optimizations for handling wireless link characteristics.

   Cross-Layer Design: Leveraging cross-layer design principles to facilitate communication and coordination between TCP and lower-layer protocols, enabling more efficient adaptation to wireless link conditions without major modifications to TCP itself

7. **What benefit can TCP draw from lower layer information? How can it be used?**

   Improved Congestion Control: TCP can benefit from lower layer information, such as packet loss and delay measurements, to enhance its congestion control algorithms. This information can be used to adjust TCP's congestion window and retransmission behavior more accurately, improving overall network efficiency and performance.

Beyond TCP:

1. **What are the advantages of moving congestion control mechanisms into the user space?**

   Flexibility: By implementing congestion control mechanisms in user space, applications have more flexibility to customize and optimize congestion control algorithms based on their specific requirements and network conditions.

   Isolation: Moving congestion control to user space allows applications to have better control over congestion control behavior without affecting the underlying operating system or other applications sharing the same network stack

Experimentation: Developers can easily experiment with different congestion control algorithms and parameters in user space without needing to modify or recompile the kernel, enabling faster iteration and innovation in congestion control research.

2. **What does HoL blocking mean?**

   HoL (Head-of-Line) Blocking refers to a situation where a packet at the head of a transmission queue is delayed or blocked, causing subsequent packets in the queue to also be delayed, even if they are ready for transmission. This can lead to increased latency and reduced throughput, especially in network protocols with strict ordering requirements, such as TCP.

3. **What is the benefit of parallel multiple data streams?**

   The benefit of parallel multiple data streams is increased throughput and reduced latency. By splitting data transmission into multiple parallel streams, applications can better utilize available network bandwidth and mitigate the effects of packet loss or delay on individual streams. This approach can improve overall performance, especially in high-speed or high-latency networks.

4. **What is a bufferbloat?**

   Bufferbloat refers to the excessive buffering of data packets in network buffers, leading to increased latency, jitter, and degraded network performance. It occurs when network buffers are oversized relative to the available bandwidth, causing delays in packet transmission and unnecessarily prolonging queueing times.

5. **What are incast issues?**

   Incast issues occur in data center networks when multiple client nodes simultaneously request data from a single server node, resulting in congestion at the server's uplink and potential packet loss. Incast issues can lead to performance degradation, increased latency, and reduced throughput, especially in networks with limited bandwidth or inadequate congestion control mechanisms

6. **What are the benefits of SPDY?**

   SPDY (pronounced "speedy") is a protocol developed by Google to improve web page load times and enhance security. Its benefits include reduced latency through multiplexing and compression of HTTP requests, decreased page load times through header compression and prioritization of requests, and enhanced security through support for SSL encryption and server push capabilities

7. **What are the benefits of QUIC?**

   QUIC (Quick UDP Internet Connections) is a transport layer protocol developed by Google that aims to improve web performance and security. Its benefits include reduced latency through multiplexing and improved congestion control algorithms, enhanced security through built-in encryption and authentication, and better reliability through mechanisms for retransmission and loss recovery

8. **What are the benefits of MPTCP?**

   MPTCP (Multipath TCP) is an extension of TCP that enables data transmission over multiple network paths simultaneously. Its benefits include improved reliability through redundancy, increased throughput by utilizing all available network paths, and enhanced resilience to network failures through dynamic path selection and failover mechanisms.